INTERNSHIP PROJECT PRESENTATION

Migration of UI Testing: Legacy to Open-Source Playwright

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ORGANIZATION: WELLS FARGO (On-Campus Offer)

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PRESENTATION AGENDA

Problem & Objective

Articulating legacy pain points and migration goals.

Technical Evolution

Accelerator, Codegen, and Playwright MCP integration.

Evaluation

Scientific principles, metrics, and self-learning.

PROBLEM & OBJECTIVES

The Challenge: Legacy UI testing scripts were brittle, slow, and expensive to maintain within the banking ecosystem.

- Frequent "Flakiness" leading to false negatives.
- Synchronous execution causing pipeline bottlenecks.
- Vendor lock-in and high licensing costs.

Objective: Modernize to Playwright for resilience, cross-browser compatibility, and speed.

TECHNICAL MIGRATION: PHASE I & II

Phase I: Accelerator

Utilised a custom **Conversion Accelerator** to translate legacy scripts into Playwright boilerplate.

Phase II: Codegen

Optimized migration using **Playwright Codegen**. This allowed for rapid generation of modern locators by recording real-time user interactions.

"Automation of the migration process reduced manual coding by approximately 65%."

PHASE III: PLAYWRIGHT MCP

Integration of the **Model Context Protocol (MCP)** allows Large Language Models to interact with the UI through structured accessibility snapshots.

- ✓ Self-healing test scripts.
- ✓ Context-aware DOM interaction.
- ✓ No vision models required; pure data-driven navigation.

TECHNOLOGY STACK

Playwright & JS

Core framework using JS for type-safety and modern async/await patterns.

MCP Server

Hosting agentic loops for autonomous exploratory testing and bug discovery.

CI/CD Integration

Automated test triggers with full trace viewer reports.

TECHNOLOGICAL COMPARISON

Feature	Legacy Framework	Playwright (New)
Execution Speed	Slow	Asynchronous & Parallel
Auto-Waiting	Manual Sleep/Implicit Waits	Native Intelligent Auto-Wait
Debugging	Console Logs / Screenshots	Trace Viewer / Time-travel
Stability	Low (Brittle Selectors)	High (Resilient Locators)

SCIENTIFIC PRINCIPLES APPLIED

- **Page Object Model (POM):** Decoupling test logic from UI selectors to ensure modularity and high maintainability.
- **The Test Pyramid Principle:** Focusing on high-value E2E journeys while maintaining isolation between test layers.
- **Atomic Testing:** Ensuring each test case is independent, allowing for infinite scalability in parallel workers.

QUALITY & QUANTUM OF WORK

40%

Reduction in Execution Time

Performance Impact

By leveraging Playwright's native parallelism, we optimized the CI/CD pipeline speedup ratio.

Speedup Estimation:

$$S = \frac{T_{legacy}}{T_{playwright}}$$

CONTINUOUS ENGAGEMENT

Month 1
Requirement gathering and Legacy Audit.

Month 2
Utilisation of Conversion Accelerator.

Month 3
Core Migration & MCP Integration.

Month 4-5
Final Validation and Knowledge Transfer.

Consistently engaged in daily Agile stand-ups and peer-review sessions.

SELF-EVALUATION & LEARNING

Personal Growth

- **Communication:** Significant improvement in articulating technical hurdles to stakeholders.
- **Corporate Grooming:** Adaptation to Wells Fargo’s work ethics and professional standards.
- **Technical Agility:** Rapidly upskilled from basic scripting to advanced agentic automation.

ANY QUESTIONS?

Wells Fargo Internship

Presentation